



Barker
College

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Student Number

2020

**YEAR 12
TRIAL HIGHER SCHOOL CERTIFICATE**

Mathematics Extension 2

Staff Involved:

WMD* ARM

40 copies

**8:30 AM
FRIDAY 14 AUGUST**

TOPICS COVERED

- **Complex Numbers I & II**
- **Proof**
- **Integration**
- **Vectors**
- **Mechanics**

General

Instructions:

- Reading time - 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

**Total marks:
100**

Section I – 10 marks (pages 2-6)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7-13)

- Attempt Questions 11-16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1–10.

1. If $\underline{a} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ then a non-zero vector $\underline{c}$ such that $\underline{a} \cdot \underline{c} = \underline{b} \cdot \underline{c} = 0$ could be

A. $\begin{pmatrix} -1 \\ 5 \\ -3 \end{pmatrix}$

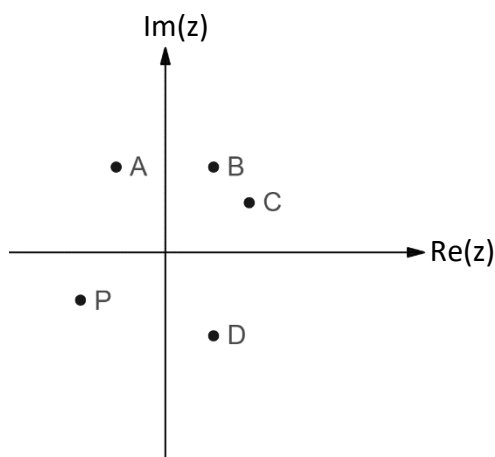
B. $\begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 5 \\ 3 \end{pmatrix}$

D. $\begin{pmatrix} -1 \\ -5 \\ -3 \end{pmatrix}$

2. A certain complex number $\frac{\bar{z}}{i}$ is represented by the point P on the Argand diagram below.

The axes have the same scale.



The complex number z is best represented by

A. A

B. B

C. C

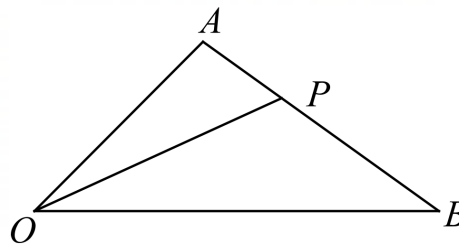
D. D

Section I continues on the next page

Section I continued

3. In this diagram, $\overrightarrow{OA} = \begin{pmatrix} 6 \\ -1 \\ 8 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} -3 \\ 4 \\ -2 \end{pmatrix}$ and $AP:PB = 1:2$.

The vector $\overrightarrow{OP}$ is equal to



A. $\frac{1}{3} \begin{pmatrix} 0 \\ 7 \\ 4 \end{pmatrix}$

B. $\begin{pmatrix} 9 \\ 7 \\ 4 \end{pmatrix}$

C. $\begin{pmatrix} 9 \\ 12 \\ 4 \end{pmatrix}$

D. $\frac{1}{3} \begin{pmatrix} 9 \\ 2 \\ 14 \end{pmatrix}$

4. If $m = a \operatorname{cis}(\theta_1)$ and $n = 3 \operatorname{cis}(\theta_2)$ and $mn = \frac{1}{2} \operatorname{cis}\left(-\frac{7\pi}{12}\right)$ then a , θ_1 and θ_2 respectively could be

A. $\frac{1}{6}, -\frac{2\pi}{3}, \frac{3\pi}{4}$

B. $6, \frac{2\pi}{3}, \frac{3\pi}{4}$

C. $\frac{1}{6}, \frac{2\pi}{3}, \frac{3\pi}{4}$

D. $6, -\frac{\pi}{3}, \frac{\pi}{4}$

Section I continues on the next page

Section I continued

5. The points R , S and T are collinear.

Given that $\overrightarrow{OR} = \underline{i} + \underline{j}$, $\overrightarrow{OS} = 2\underline{i} - \underline{j} + \underline{k}$ and $\overrightarrow{OT} = 3\underline{i} + m\underline{j} + n\underline{k}$, the values of m and n are

A. $m = 3, n = -2$

B. $m = -1, n = 0$

C. $m = 0, n = 1$

D. $m = -3, n = 2$

6. Consider the complex numbers $2z$, $-iz$ and $2z - iz$, where $z \neq 0$.

These three complex numbers are plotted in the Argand plane and together with the origin O , they form the vertices of a quadrilateral.

The area of this quadrilateral is

A. $2|z^2|$

B. $|2z|$

C. $|z^2|$

D. $|z| + |2z|$

Section I continues on the next page

Section I continued

7. Given the vector $\underline{a} = \frac{1}{2}(\sqrt{2}\underline{i} - \underline{j} + \underline{k})$, then the vector $\underline{a}$
- A. makes an angle of 120° with the positive y -axis and 30° with the positive z -axis.
 - B. makes an angle of 45° with the positive x -axis and 120° with the positive y -axis.
 - C. makes an angle of 45° with the positive x -axis and 150° with the positive y -axis.
 - D. makes an angle of 135° with the positive x -axis and 150° with the positive y -axis.
8. The number of distinct roots of the equation $(z^2 - 2zi - 1)(z^2 + 2i)(z^2 + 2zi + 2) = 0$ is
- A. 3
 - B. 4
 - C. 5
 - D. 6

Section I continues on the next page

Section I continued

9. The triangle formed by the three points whose position vectors are

$$\begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} \text{ is}$$

- A. a scalene triangle
- B. a right-angled triangle which is not isosceles
- C. an isosceles triangle which is not right-angled
- D. a right-angled isosceles triangle
10. By using the substitution $u = \cos x$, or otherwise, it can be shown that $\int \cos^2 x \sin^7 x \, dx =$

- A. $-\frac{\cos^3 x}{3} + \frac{3\cos^5 x}{5} - \frac{3\cos^7 x}{7} + \frac{\cos^9 x}{9} + C$
- B. $-\cos^3 x + 3\cos^5 x - 3\cos^7 x + \cos^9 x + C$
- C. $\frac{\cos^3 x}{3} - \frac{3\cos^5 x}{5} + \frac{3\cos^7 x}{7} - \frac{\cos^9 x}{9} + C$
- D. $\cos^3 x - 3\cos^5 x + 3\cos^7 x - \cos^9 x + C$

End of Section I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Let $z_1 = -1 + 3i$ and $z_2 = 2 - i$.

Find the following, in simplest form:

(i) $z_1 + z_2$ 1

(ii) $z_1 z_2$ 1

(iii) $\operatorname{Re}(z_1) - \operatorname{Im}(z_2)$ 1

(b) (i) Find $\frac{d}{dx}(\log_e(x^2 + 1))$. 1

(ii) Evaluate $\int_0^{\sqrt{e^2-1}} \frac{4x}{x^2+1} \log_e(x^2+1) dx$, answering in simplified form. 2

(c) (i) Express $5 \cos \theta - 2 \sin \theta$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2

(ii) Hence, or otherwise, find $\int \frac{dx}{\left(5 \cos \frac{x}{2} - 2 \sin \frac{x}{2}\right)^2}$. 3

(d) A complex number z satisfies the inequality $|z + 2 - 2\sqrt{3}i| \leq 2$.

(i) Sketch the corresponding region representing all possible values of z . 2

(ii) Find the set of possible values $\operatorname{Arg} z$ can take. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Simplify $(1-i)^6$. 1
- (ii) Solve the equation $z^3 = (1-i)^6$, writing the solutions in the form $x+iy$. 2
- (b) The complex number $3-i$ is denoted by u .
- (i) Express $\frac{\bar{u}}{u}$ in the form $x+iy$. 1
- (ii) By considering the argument of $\frac{\bar{u}}{u}$, prove that $\tan^{-1}\left(\frac{3}{4}\right) = 2 \tan^{-1}\left(\frac{1}{3}\right)$. 2
- (c) What are the values of real numbers p and q such that $1-i$ is a root of the equation $z^3 + pz + q = 0$? 2
- (d) (i) By using the method of completing the square, or otherwise, 2
solve $z^2 - 2z \cos \theta + 1 = 0$. Give your simplified solutions for z in terms of θ .
- (ii) Let α and β be the two solutions found in (i). If P and Q are points 2
on the Argand diagram representing $\alpha^n + \beta^n$ and $\alpha^n - \beta^n$ respectively,
show that PQ is of constant length for n , $n \in \mathbf{Z}^+$.
- (e) Find the coordinates of the point which is nearest to the origin on the line 3
- $$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix}.$$

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) Using the exponential form of a complex number, prove that

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta), \quad n \in \mathbf{Z}. \quad \mathbf{1}$$

- (b) (i) Prove that a possible value of $(-1)^{-i}$ is e^π . **1**

- (ii) Hence, or otherwise, find an expression for all the possible values of $(-1)^{-i}$. **1**

- (c) A particle's motion is defined by the equation $v^2 = 12 + 4x - x^2$, where x is its displacement from the origin in metres and v its velocity in ms^{-1} .

Initially the particle is 6 metres to the right of the origin.

- (i) Show that the particle is moving in Simple Harmonic Motion, stating the centre of motion. **2**

- (ii) Find the period and the amplitude of the motion. **2**

- (iii) The displacement of the particle at any time t is given by the equation **2**

$$x = a \sin(nt + \theta) + b.$$

Find the values of θ and b , given $0 \leq \theta < 2\pi$.

- (d) A particle of mass m kg is fired directly upwards with speed 200 ms^{-1} in a medium where the resistance is $\frac{1}{10}mv$ newtons when the speed is $v \text{ ms}^{-1}$. Let $g = 10 \text{ ms}^{-2}$.

Hence, for the upward journey, $\ddot{x} = -\frac{1}{10}(100 + v)$, where x metres is the vertical displacement from the point of projection.

- (i) Show that the maximum height attained above the point of projection is $1000(2 - \log_e 3)$ metres. **3**

- (ii) Show that the speed $v \text{ ms}^{-1}$ of the particle on return to its point of projection **3**

$$\text{satisfies the equation } \frac{v}{100} + \ln \left| 1 - \frac{v}{100} \right| + (2 - \ln 3) = 0.$$

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, **2**
The square numbers are 1, 4, 9, 16, 25, 36, 49, 64,
With one exception, if a square number differs from a prime by 1, the prime is bigger.
For example, 16 is a square and 17 is a prime. The prime is bigger. The only exception is prime 3 and square 4.
It is conjectured that for all consecutive numbers that are a prime and a square, except for 3 and 4, the prime is bigger.
Prove the conjecture is true.
- (b) Use proof by contradiction to prove that if a, b are integers, then $a^2 - 4b - 3 \neq 0$. **3**
You may wish to consider the case when a is even and the case when a is odd.
- (c) A sequence of numbers is given by $T_1 = 6$, $T_2 = 27$ and $T_n = 6T_{n-1} - 9T_{n-2}$ **3**
for integers $n \geq 3$.
Use mathematical induction to show that $T_n = (n+1)3^n$ for integers $n \geq 1$.
- (d) (i) Use De Moivre's theorem to show that $(1 + i \tan \theta)^5 = \frac{\cos 5\theta + i \sin 5\theta}{\cos^5 \theta}$. **1**
- (ii) Hence find expressions for $\cos 5\theta$ and $\sin 5\theta$ in terms of $\tan \theta$ and $\cos \theta$. **2**
- (iii) Show that $\tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$ where $t = \tan \theta$. **1**
- (iv) Use the result of (iii) and an appropriate substitution **3**
to show that $\tan \frac{\pi}{5} = \sqrt{5 - 2\sqrt{5}}$.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The trajectory of a projectile fired with speed $u \text{ ms}^{-1}$ at an angle θ to the horizontal, in a medium whose resistance to the projectile's motion is proportional to the projectile's velocity, is represented by the parametric equations

$$x = \frac{u \cos \theta}{k} (1 - e^{-kt}) \quad \text{and} \quad y = \frac{(10 + ku \sin \theta)}{k^2} (1 - e^{-kt}) - \frac{10t}{k},$$

where k is the constant of proportionality of the resistance.

- (i) Show the greatest height is reached when $t = \frac{1}{k} \log_e \left(\frac{10 + ku \sin \theta}{10} \right)$. **2**

- (ii) If $k = 0.5$, $u = 40$ and $\theta = 30^\circ$, show that the greatest height is reached when the projectile is at the point $(20\sqrt{3}, 40(1 - \log_e 2))$. **2**

(b) $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx, \quad n = 1, 2, 3, \dots$

- (i) Show that $I_{n+1} = \frac{2n-1}{2n} I_n + \frac{1}{(2^{n+1})n}$. **3**

- (ii) Hence evaluate $\int_0^1 \frac{1}{(1+x^2)^3} dx$. **2**

Question 15 continues on the next page

Question 15 continued

- (c) Relative to a fixed origin O , the points A , B and C have position vectors

$$\underline{a} = \begin{pmatrix} -1 \\ 4 \\ 3 \\ 7 \end{pmatrix}, \underline{b} = \begin{pmatrix} 4 \\ 4 \\ 3 \\ 2 \end{pmatrix} \text{ and } \underline{c} = \begin{pmatrix} 6 \\ 16 \\ 3 \\ 2 \end{pmatrix} \text{ respectively.}$$

- (i) Find the cosine of $\angle ABC$. **1**
- (ii) Hence find the area of the triangle ABC . **1**
- (iii) Use a vector method to find the shortest distance between the point A and the line passing through the points B and C . **2**

Let D be the point such that the quadrilateral $ABCD$ is a kite, where $BC = CD$ and $BA = AD$.

- (iv) Find the position vector of the point D . **2**

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Use the substitution $t = \tan \frac{x}{2}$ to show $\int_0^{\frac{\pi}{2}} \frac{dx}{2 - \cos x + 2 \sin x} = \log_e 2$. **4**

(b) (i) Write down an expression for the sum of the arithmetic series **1**

$$1 + 2 + 3 + 4 + \dots + (n-1).$$

(ii) Using calculus, or otherwise, show that for $x > 0$, $x > \log_e(1+x)$. **2**

(iii) Using (i) and (ii), show that $e^{\binom{n}{2}} > n!$ for positive integers $n = 2, 3, 4, \dots$ **3**

(c) Consider the function $f(x) = \sum_{k=1}^n \left(\sqrt{a_k} x - \frac{1}{\sqrt{a_k}} \right)^2$ where $a_1, a_2, \dots, a_n$ are positive real numbers.

(i) By expressing $f(x)$ as a quadratic function of x , show that **3**

$$(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \geq n^2.$$

(ii) Hence show that $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \geq \frac{2n}{n+1}$. **2**

End of Question 16

End of Paper

YEAR 12 EXTENSION 2 TRIAL EXAMINATION 2020

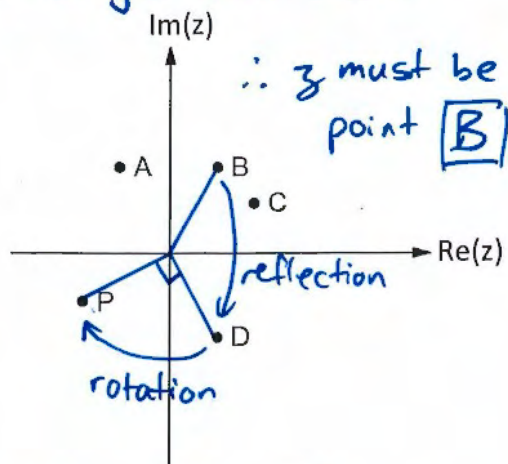
1. Test each answer.

$$\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 5 \\ 3 \end{pmatrix} = -2 + 5 - 3 = 0$$

$$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 5 \\ 3 \end{pmatrix} = -1 - 5 + 6 = 0$$

C

2. To produce $\frac{\bar{z}}{i}$, z has been reflected in the real axis then rotated $\frac{\pi}{2}$ about the origin clockwise.



$$\begin{aligned} 3. \vec{OP} &= \vec{OA} + \vec{AP} \\ &= \vec{OA} + \frac{1}{3}\vec{AB} \\ &= \vec{OA} + \frac{1}{3}(\vec{OB} - \vec{OA}) \\ &= \begin{pmatrix} 6 \\ -1 \\ 8 \end{pmatrix} + \frac{1}{3} \left[\begin{pmatrix} -3 \\ 4 \\ -2 \end{pmatrix} - \begin{pmatrix} 6 \\ -1 \\ 8 \end{pmatrix} \right] \end{aligned}$$

$$= \begin{pmatrix} 6 \\ -1 \\ 8 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} -9 \\ 5 \\ -10 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ \frac{2}{3} \\ \frac{14}{3} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 9 \\ 2 \\ 14 \end{pmatrix}$$

D

$$4. mn = 3a \operatorname{cis}(\theta_1 + \theta_2)$$

$$\therefore \frac{1}{2} = 3a \quad \text{so} \quad a = \frac{1}{6} \quad \therefore \text{A or C}$$

$$\text{Try } \frac{-2\pi}{3} + \frac{3\pi}{4} = \frac{\pi}{12} \neq \frac{-7\pi}{12} \quad (\text{not A})$$

$$\text{Try } \frac{2\pi}{3} + \frac{3\pi}{4} = \frac{17\pi}{12}$$

$$\frac{17\pi}{12} - 2\pi = \frac{-7\pi}{12}$$

C

5. Since collinear, $\vec{RS} = \lambda \vec{ST}$

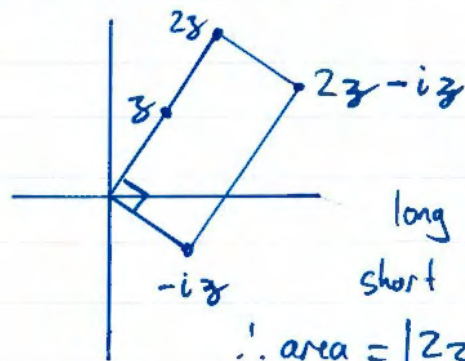
$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \lambda \left[\begin{pmatrix} 3 \\ m \\ n \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right]$$

$$\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ m+1 \\ n-1 \end{pmatrix} \quad \text{so } \lambda = 1$$

$$\begin{aligned} -2 &= m+1 & 1 &= n-1 \\ m &= -3 & n &= 2 \end{aligned}$$

D

6.



long side length = $|2z|$

short side length = $|z|$

$$\therefore \text{area} = |2z||z| = 2|z|^2 \quad \text{A}$$

$$7. \underline{a} = \frac{1}{\sqrt{2}}\underline{i} - \frac{1}{2}\underline{j} + \frac{1}{2}\underline{k}$$

$$\cos \theta_x = \frac{1}{\sqrt{2}} \quad \cos \theta_y = -\frac{1}{2} \quad \cos \theta_z = \frac{1}{2}$$

$$\theta_x = 45^\circ \quad \theta_y = 120^\circ \quad \theta_z = 60^\circ$$

B

8. Factorise to:

$$(z-i)^2(z^2+2i)(z^2+2zi+2)=0$$

Degree 6 polynomial with one double root $\Rightarrow$ 5 distinct roots.

C

9. Vectors for the three sides:

$$\begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \left| \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \right| = 3$$

$$\begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -4 \end{pmatrix}, \left| \begin{pmatrix} -1 \\ 1 \\ -4 \end{pmatrix} \right| = 3\sqrt{2}$$

$$\begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}, \left| \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \right| = 3$$

$\therefore$ isosceles

$$\text{And since } 3^2 + 3^2 = (3\sqrt{2})^2$$

$$\left(\text{or since } \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} = 0 \right) \text{ also}$$

a right angle triangle. D

$$10. \frac{du}{dx} = -\sin x \Rightarrow dx = \frac{-du}{\sin x}$$

$$\begin{aligned} I &= \int u^2 (1-u^2)^3 \sin x \cdot \frac{-du}{\sin x} \\ &= -\int u^2 (1-3u^2+3u^4-u^6) du \\ &= -\left(\frac{u^3}{3} - 3\frac{u^5}{5} + 3\frac{u^7}{7} - \frac{u^9}{9} \right) + C \\ &= -\frac{u^3}{3} + \frac{3u^5}{5} - \frac{3u^7}{7} + \frac{u^9}{9} + C \end{aligned}$$

A

Question 11.

$$a) i) -1+3i+2-i = 1+2i$$

$$\begin{aligned} ii) (-1+3i)(2-i) &= -2+i+6i-3i^2 \\ &= -2+3+7i \\ &= 1+7i \end{aligned}$$

$$iii) -1-(-1) = 0$$

$$b) i) \frac{2x}{x^2+1}$$

$$ii) \int_0^{\sqrt{e^2-1}} 2 \cdot \frac{2x}{x^2+1} \cdot \log_e(x^2+1) dx$$

$$= \left[2 \cdot \frac{1}{2} (\log_e(x^2+1))^2 \right]_0^{\sqrt{e^2-1}}$$

$$= (\log_e(e^2-1+1))^2 - (\log_e(0+1))^2$$

$$= (\log_e(e^2))^2 - 0$$

$$= 4$$

$$c) i) 5 \cos \theta - 2 \sin \theta = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$5 = R \cos \alpha \quad R = \sqrt{5^2 + 2^2}$$

$$2 = R \sin \alpha \quad = \sqrt{29}$$

$$\frac{2}{5} = \tan \alpha \Rightarrow \alpha = \tan^{-1} \frac{2}{5}$$

$$\therefore 5 \cos \theta - 2 \sin \theta = \sqrt{29} \cos \left(\theta + \tan^{-1} \frac{2}{5} \right)$$

$$ii) I = \int \frac{dx}{\left(\sqrt{29} \cos \left(\frac{x}{2} + \alpha \right) \right)^2} \text{ where } \alpha = \tan^{-1} \frac{2}{5}$$

$$= \frac{1}{29} \int \sec^2 \left(\frac{x}{2} + \alpha \right) dx$$

$$= \frac{1}{29} \cdot 2 \tan \left(\frac{x}{2} + \alpha \right) + C$$

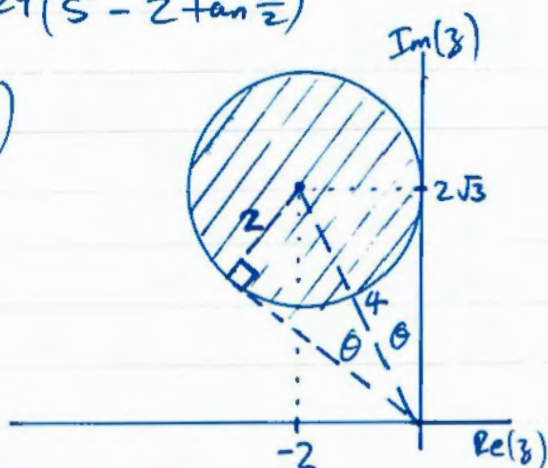
$$= \frac{2}{29} \tan \left(\frac{x}{2} + \tan^{-1} \frac{2}{5} \right) + C$$

or, alternatively:

$$= \frac{2}{29} \frac{\tan \frac{x}{2} + \frac{2}{5}}{1 - (\tan \frac{x}{2})(\frac{2}{5})}$$

$$= \frac{2(5 \tan \frac{x}{2} + 2)}{29(5 - 2 \tan \frac{x}{2})}$$

d) i)



Circle, radius 2, centre $-2 + 2\sqrt{3}i$

ii) Minimum value for $\text{Arg}(z) = \frac{\pi}{2}$
 $|-2 + 2\sqrt{3}i| = \sqrt{4 + (2\sqrt{3})^2} = \sqrt{16} = 4$

so $\sin \theta = \frac{2}{4}$ and $\theta = \frac{\pi}{6}$

Max value for $\text{Arg}(z) = \frac{\pi}{2} + 2 \times \frac{\pi}{6} = \frac{5\pi}{6}$

$\therefore \frac{\pi}{2} \leq \text{Arg}(z) \leq \frac{5\pi}{6}$

Question 12

a) i) $(1-i)^6 = (\sqrt{2} e^{-\frac{\pi}{4}i})^6$
 $= 8 e^{-\frac{6\pi}{4}i}$
 $= 8 e^{\frac{\pi}{2}i} = 8i$

ii) $z^3 = 8i$ let $z = r \text{cis } \theta$

$r^3 \text{cis } 3\theta = 8 \text{cis } (\frac{\pi}{2} + 2k\pi)$

$r^3 = 8 \Rightarrow r = 2$

$3\theta = \frac{\pi}{2} + 2k\pi$

$\theta = \frac{\pi(1+4k)}{6}$

and $-\pi < \frac{\pi(1+4k)}{6} \leq \pi$

$-6 < 1+4k \leq 6$

$-\frac{7}{4} < k \leq \frac{5}{4}$

So, for $k = -1, 0, 1$

$\theta = -\frac{3\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$

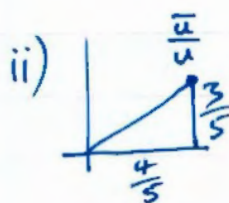
$z = 2 \text{cis } (-\frac{\pi}{2}), 2 \text{cis } (\frac{\pi}{6}), 2 \text{cis } (\frac{5\pi}{6})$

$= -2i, \sqrt{3} + i, -\sqrt{3} + i$

b) i) $\frac{3+i}{3-i} \times \frac{3+i}{3+i} = \frac{9+6i+i^2}{9+1}$

$= \frac{8+6i}{10}$

$= \frac{4}{5} + \frac{3}{5}i$



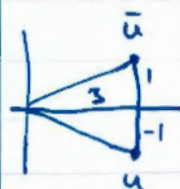
ii) $\arg(\frac{\bar{u}}{u}) = \tan^{-1}(\frac{3/5}{4/5})$
 $= \tan^{-1}(\frac{3}{4}) \quad \text{--- (1)}$

But $\arg(\frac{\bar{u}}{u}) = \arg \bar{u} - \arg u$

$= \tan^{-1}(\frac{1}{3}) - \tan^{-1}(\frac{-1}{3})$

$= \tan^{-1}(\frac{1}{3}) + \tan^{-1}(\frac{1}{3})$

$= 2 \tan^{-1}(\frac{1}{3}) \quad \text{--- (2)}$



From (1) and (2):

$\tan^{-1}(\frac{3}{4}) = 2 \tan^{-1}(\frac{1}{3})$

(4)

c) If $1-i$ is a root,

$$(1-i)^3 + p(1-i) + q = 0$$

$$-2-2i + p - pi + q = 0$$

$$(-2+p+q) + (-2-p)i = 0$$

$$-2-p=0 \quad -2+p+q=0$$

$$p=-2 \quad -4+q=0$$

$$q=4$$

d) i)

$$z^2 - 2z \cos \theta + \cos^2 \theta = 1 + \cos^2 \theta$$

$$(z - \cos \theta)^2 = -\sin^2 \theta$$

$$z - \cos \theta = \pm i \sin \theta$$

$$z = \cos \theta \pm i \sin \theta$$

ii) $\alpha = \cos \theta + i \sin \theta$

$$\alpha^n = \cos(n\theta) + i \sin(n\theta)$$

$$\beta = \cos \theta - i \sin \theta$$

$$= \cos(-\theta) + i \sin(-\theta)$$

$$\beta^n = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos(n\theta) - i \sin(n\theta)$$

$$\alpha^n + \beta^n = 2 \cos(n\theta)$$

$$\alpha^n - \beta^n = 2i \sin(n\theta)$$

$$PQ = |(\alpha^n + \beta^n) - (\alpha^n - \beta^n)|$$

$$= |2 \cos(n\theta) - 2i \sin(n\theta)|$$

$$= 2 \sqrt{\cos^2(n\theta) + \sin^2(n\theta)}$$

$$= 2\sqrt{1}$$

$$= 2$$

e) $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1-\lambda \\ 2-3\lambda \\ 2 \end{pmatrix}$

Distance from O to a point on the line is:

$$d = \sqrt{x^2 + y^2 + z^2}$$

$$d^2 = (1-\lambda)^2 + (2-3\lambda)^2 + 2^2$$

$$= 1 - 2\lambda + \lambda^2 + 4 - 12\lambda + 9\lambda^2 + 4$$

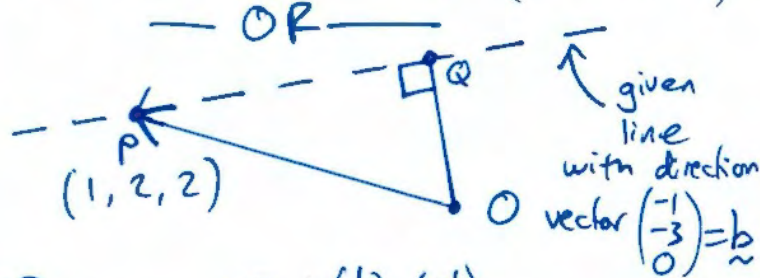
$$= 10\lambda^2 - 14\lambda + 9$$

minimum value of d^2 (and hence d) is when $\lambda = \frac{-b}{2a}$

$$= \frac{14}{20} = \frac{7}{10}$$

So, closest point to O is:

$$\left(1 - \frac{7}{10}, 2 - \frac{3 \times 7}{10}, 2\right) = \left(\frac{3}{10}, \frac{-1}{10}, 2\right)$$



$$\begin{aligned} \vec{QP} &= \text{proj}_{\vec{b}} \vec{OP} = \frac{\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix}}{\begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix}} \begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix} \\ &= \frac{-7}{10} \begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{7}{10} \\ \frac{21}{10} \\ 0 \end{pmatrix} \end{aligned}$$

$$\vec{OQ} = \vec{OP} - \vec{QP} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} \frac{7}{10} \\ \frac{21}{10} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{10} \\ -\frac{1}{10} \\ 2 \end{pmatrix}$$

Question 13

a) $\cos \theta + i \sin \theta = e^{i\theta}$
 $(\cos \theta + i \sin \theta)^n = (e^{i\theta})^n$
 $= e^{in\theta}$
 $= \cos(n\theta) + i \sin(n\theta)$

b) i) Euler's identity:

$$e^{i\pi} = -1$$

$$(e^{i\pi})^{-i} = (-1)^{-i}$$

$$e^{-i^2\pi} = (-1)^{-i}$$

$$e^{\pi} = (-1)^{-i}$$

ii) Similarly:

$$-1 = e^{i(\pi + 2k\pi)}, k \in \mathbb{Z}$$

$$(-1)^{-i} = e^{-i^2(\pi + 2k\pi)}$$

$$= e^{(2k+1)\pi}$$

c) i) $\frac{1}{2}v^2 = 6 + 2x - \frac{1}{2}x^2$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 2 - x$$

$$\ddot{x} = -1^2(x - 2)$$

which is of the form

$$\ddot{x} = -n^2(x - c)$$

$\therefore$ SHM with centre 2.

ii) Min/max x at $v=0$:

$$0 = 12 + 4x - x^2$$

$$0 = (x-6)(x+2)$$

$$x = -2, 6$$

$$\text{amplitude} = 6 - 2 = 2 - (-2) = 4$$

period = $\frac{2\pi}{n} = \frac{2\pi}{1} = 2\pi$

(iii) centre of motion, $b = 2$

$$x = 4 \sin(t + \theta) + 2$$

at $t=0$, $x=6$

$$6 = 4 \sin(0 + \theta) + 2$$

$$1 = \sin \theta$$

$$\theta = \frac{\pi}{2}$$

d) i)

$$v \frac{dv}{dx} = \frac{-1}{10}(100 + v)$$

$$\int_{200}^0 \frac{v dv}{100 + v} = \frac{-1}{10} \int_0^h dx$$

$$\int_{200}^0 \frac{100 + v - 100}{100 + v} dv = \frac{-1}{10} \left[x \right]_0^h$$

$$\int_{200}^0 \left(1 - \frac{100}{100 + v} \right) dv = \frac{-h}{10}$$

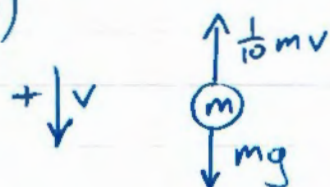
$$\left[v - 100 \ln(100 + v) \right]_{200}^0 = \frac{-h}{10}$$

$$-100 \ln 100 - 200 + 100 \ln 300 = \frac{-h}{10}$$

$$-2000 + 100 \ln\left(\frac{300}{100}\right) = -h$$

$$h = 1000(2 - \ln 3)$$

ii)



Let max height be $x=0$ now. ($v=0$ there)

$$ma = F$$

$$ma = mg - \frac{1}{10}mv$$

$$v \frac{dv}{dx} = 10 - \frac{v}{10}$$

(6)

$$v \frac{dv}{dx} = \frac{100-v}{10}$$

$$\int_0^v \frac{v dv}{100-v} = \frac{1}{10} \int_0^{1000(2-\ln 3)} dx$$

$$-\int_0^v \frac{v-100+100}{v-100} dv = \frac{1}{10} [x]_0^{1000(2-\ln 3)}$$

$$\int_0^v \left(1 + \frac{100}{v-100}\right) dv = -\frac{1}{10} (1000(2-\ln 3))$$

$$\left[v + 100 \ln |v-100| \right]_0^v = -100(2-\ln 3)$$

$$v + 100 \ln |v-100| - 0 - 100 \ln |0-100| + 100(2-\ln 3) = 0$$

$$\frac{v}{100} + \ln \left| \frac{v-100}{100} \right| + (2-\ln 3) = 0$$

$$\frac{v}{100} + \ln \left| \frac{v}{100} - 1 \right| + (2-\ln 3) = 0$$

$$\frac{v}{100} + \ln \left| 1 - \frac{v}{100} \right| + (2-\ln 3) = 0$$

Question 14

a) Let the prime be p and the square number be n^2 .

For $n^2=1$ and $p=2$, $p > n^2$.

For $n^2=4$ and $p=3$, $p < n^2$ (the exception)

For $n \geq 3$, if $p = n^2 - 1$ then $p = (n-1)(n+1)$ where, since $n \geq 3$, $n-1 > 1$ and $n+1 > 1$ which implies that p is composite.

This is a contradiction so the situation where the prime is one less than the square is impossible. Hence, for $n \geq 3$, $p = n^2 + 1$.

$\therefore$ The conjecture is true.

b) Assume $a^2 - 4b - 3 = 0$ ($a, b \in \mathbb{Z}$)

$$\text{so } a^2 - 4b = 3$$

If a is even, $a = 2n$ ($n \in \mathbb{Z}$)

$$(2n)^2 - 4b = 3$$

$$4n^2 - 4b = 3$$

$$2(2n^2 - 2b) = 3$$

which is a contradiction since LHS is even.

If a is odd, $a = 2n+1$ ($n \in \mathbb{Z}$)

$$(2n+1)^2 - 4b = 3$$

$$4n^2 + 4n - 4b = 2$$

$$4(n^2 + n - b) = 2$$

$$n^2 + n - b = \frac{1}{2}$$

which is a contradiction since LHS must be an integer

$\therefore a, b \in \mathbb{Z} \Rightarrow a^2 - 4b - 3 \neq 0$

c) Prove for $n=1$:

$$T_1 = (1+1)3^1$$

$$= 2 \times 3$$

$$= 6 \quad \text{as stated}$$

Prove for $n=2$:

$$T_2 = (2+1)3^2$$

$$= 3 \times 9$$

$$= 27 \quad \text{as stated}$$

(7)

Assume true for $n=k$, $k \geq 1$

i.e. $T_k = (k+1)3^k$

Assume true for $n=k+1$, $k \geq 1$

i.e. $T_{k+1} = (k+1+1)3^{k+1}$
 $= (k+2)3^{k+1}$

Now prove for $n=k+2$, $k \geq 1$
 RTP: $T_{k+2} = (k+3)3^{k+2}$

$$T_{k+2} = 6T_{k+1} - 9T_k$$

$$= 6T_{k+1} - 9T_k$$

$$= 6(k+2)3^{k+1} - 9(k+1)3^k$$

$$= 2(k+2) \cdot 3 \cdot 3^{k+1} - (k+1) \cdot 3^2 \cdot 3^k$$

$$= (2k+4)3^{k+2} - (k+1)3^{k+2}$$

$$= (2k+4-k-1)3^{k+2}$$

$$= (k+3)3^{k+2}$$

Hence by the principle of mathematical induction, the statement is true $\forall n \in \mathbb{Z}$, $n \geq 1$.

d) i) $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$

$$(\cos \theta [1 + i \tan \theta])^5 = \cos 5\theta + i \sin 5\theta$$

$$\cos^5 \theta (1 + i \tan \theta)^5 = \cos 5\theta + i \sin 5\theta$$

$$(1 + i \tan \theta)^5 = \frac{\cos 5\theta + i \sin 5\theta}{\cos^5 \theta}$$

ii) From part i)

$$\text{LHS} = 1^5 + 5i \tan \theta + 10i^2 \tan^2 \theta + 10i^3 \tan^3 \theta + 5i^4 \tan^4 \theta + i^5 \tan^5 \theta$$

$$= 1 - 10 \tan^2 \theta + 5 \tan^4 \theta$$

$$+ i(5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta)$$

$$\text{RHS} = \frac{\cos 5\theta}{\cos^5 \theta} + i \frac{\sin 5\theta}{\cos^5 \theta}$$

Equating real/imaginary parts:

$$\frac{\cos 5\theta}{\cos^5 \theta} = 1 - 10 \tan^2 \theta + 5 \tan^4 \theta$$

$$\cos 5\theta = \cos^5 \theta (1 - 10 \tan^2 \theta + 5 \tan^4 \theta)$$

Similarly:

$$\sin 5\theta = \cos^5 \theta (5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta)$$

iii) $\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta}$

$$= \frac{\cos^5 \theta (5t - 10t^3 + t^5)}{\cos^5 \theta (1 - 10t^2 + 5t^4)}$$

$$= \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$$

iv) Let $\theta = \frac{\pi}{5}$ so $t = \tan \frac{\pi}{5}$

and $\tan 5\theta = \tan \pi = 0$

$$\text{so } 0 = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$$

$$0 = t(t^4 - 10t^2 + 5)$$

But $t = \tan \frac{\pi}{5} \neq 0$ so,

$$(t^2)^2 - 10t^2 + 5 = 0$$

$$t^2 = \frac{10 \pm \sqrt{100 - 4 \times 5}}{2}$$

$$= 5 \pm 2\sqrt{5}$$

but since $\tan x$ is always increasing, $\tan \frac{\pi}{5} < \tan \frac{\pi}{4}$

$$\tan \frac{\pi}{5} < 1$$

$$\text{so } t^2 < 1$$

$$t^2 = 5 - 2\sqrt{5}$$

$$t = \pm \sqrt{5 - 2\sqrt{5}}$$

$$\text{but } t = \tan \frac{\pi}{5} > 0$$

$$\therefore \tan \frac{\pi}{5} = \sqrt{5 - 2\sqrt{5}}$$

Question 15

$$\text{a) i) } \dot{y} = \frac{(10 + k u \sin \theta)}{k^2} (k e^{-kt}) - \frac{10}{k}$$

and max height is at $\dot{y} = 0$

$$\frac{(10 + k u \sin \theta) k e^{-kt}}{k^2} = \frac{10}{k}$$

$$e^{-kt} = \frac{10}{10 + k u \sin \theta}$$

$$-kt = \ln \left(\frac{10}{10 + k u \sin \theta} \right)$$

$$t = \frac{1}{k} \ln \left(\frac{10 + k u \sin \theta}{10} \right)$$

$$\text{ii) } t = \frac{1}{0.5} \ln \left(\frac{10 + 0.5 \times 40 \times \sin 30^\circ}{10} \right)$$

$$= 2 \ln 2$$

at $t = 2 \ln 2$:

$$y = \frac{10 + 0.5 \times 40 \times \sin 30^\circ}{0.5^2} (1 - e^{-0.5(2 \ln 2)}) - \frac{10(2 \ln 2)}{0.5}$$

$$= 80(1 - \frac{1}{2}) - 40 \ln 2$$

$$= 40 - 40 \ln 2$$

$$= 40(1 - \ln 2)$$

$$x = \frac{40 \cos 30^\circ}{0.5} (1 - e^{-0.5(2 \ln 2)})$$

$$= 40\sqrt{3}(1 - \frac{1}{2})$$

$$= 20\sqrt{3}$$

$\therefore$ max height at $(20\sqrt{3}, 40(1 - \ln 2))$

$$\text{b) i) } I_n = \int_0^1 (1+x^2)^{-n} dx$$

$$\begin{aligned} u &= (1+x^2)^{-n} & v' &= 1 \\ u' &= -n(1+x^2)^{-n-1} \cdot 2x & v &= x \\ &= \frac{-2nx}{(1+x^2)^{n+1}} \end{aligned}$$

$$\text{so, } I_n = \left[x(1+x^2)^{-n} \right]_0^1 - \int_0^1 \frac{-2nx^2}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n \int_0^1 \frac{1+x^2 - 1}{(1+x^2)^{n+1}} dx$$

$$= \frac{1}{2^n} + 2n \int_0^1 \frac{1}{(1+x^2)^n} dx - 2n \int_0^1 \frac{dx}{(1+x^2)^{n+1}}$$

$$I_n = \frac{1}{2^n} + 2n I_n - 2n I_{n+1}$$

(9)

$$2_n I_{n+1} = 2_n I_n - I_n + \frac{1}{2^n}$$

$$I_{n+1} = \frac{2_n I_n - I_n}{2_n} + \frac{1}{2_n \cdot 2^n}$$

$$= \frac{2_n - 1}{2_n} I_n + \frac{1}{(2^{n+1})_n}$$

$$\text{ii)} \quad I_1 = \int_0^1 \frac{1}{1+x^2} dx$$

$$= [\tan^{-1} x]_0^1$$

$$= \frac{\pi}{4}$$

$$I_2 = \frac{2-1}{2} \cdot \frac{\pi}{4} + \frac{1}{2^2}$$

$$= \frac{\pi+2}{8}$$

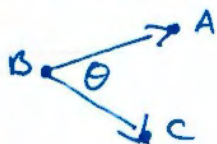
$$\int_0^1 \frac{1}{(1+x^2)^3} dx = I_3$$

$$= \frac{4-1}{4} \times \frac{\pi+2}{8} + \frac{1}{2^3 \times 2}$$

$$= \frac{3(\pi+2)}{32} + \frac{1}{16}$$

$$= \frac{3\pi+8}{32}$$

c) i) Let $\angle ABC = \theta$



$$\vec{BA} \cdot \vec{BC} = |\vec{BA}| |\vec{BC}| \cos \theta$$

$$\vec{BA} = \begin{pmatrix} -1 \\ \frac{4}{3} \\ 7 \end{pmatrix} - \begin{pmatrix} 4 \\ \frac{4}{3} \\ 2 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \\ 5 \end{pmatrix}$$

$$|\vec{BA}| = \sqrt{(-5)^2 + 5^2} = \sqrt{50}$$

$$\vec{BC} = \begin{pmatrix} 6 \\ \frac{16}{3} \\ 2 \end{pmatrix} - \begin{pmatrix} 4 \\ \frac{4}{3} \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}$$

$$|\vec{BC}| = \sqrt{2^2 + 4^2} = \sqrt{20}$$

$$\begin{pmatrix} -5 \\ 0 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \sqrt{50} \sqrt{20} \cos \theta$$

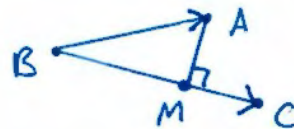
$$\cos \theta = \frac{-10}{\sqrt{1000}} = -\frac{1}{\sqrt{10}}$$

$$\text{ii)} \quad \begin{array}{c} \sqrt{10} \\ \triangle \\ 1 \end{array} \quad \sin \theta = \frac{3}{\sqrt{10}}$$

$$\text{Area } \triangle ABC = \frac{1}{2} \sqrt{50} \sqrt{20} \cdot \frac{3}{\sqrt{10}}$$

$$= 15 \text{ units}^2$$

iii)



$$\vec{BM} = \text{proj}_{\vec{BC}} \vec{BA}$$

$$= \frac{\begin{pmatrix} -5 \\ 0 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}}{\begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}$$

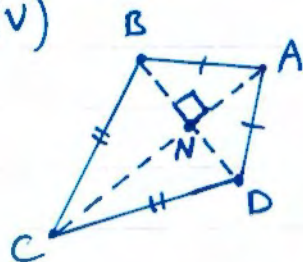
$$= \frac{-10}{20} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}$$

$$|\vec{BM}| = \sqrt{(-1)^2 + (-2)^2} = \sqrt{5}$$

$$AM = \sqrt{50^2 - 5^2} \quad (\text{Pythagoras})$$

$$= 3\sqrt{5}$$

iv)



$$\vec{BN} = \vec{BA} + \lambda \vec{AC}$$

$$= \begin{pmatrix} -5 \\ 0 \\ 5 \end{pmatrix} + \lambda \left[\begin{pmatrix} 6 \\ \frac{16}{3} \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ \frac{4}{3} \\ 7 \end{pmatrix} \right]$$

$$= \begin{pmatrix} -5 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 4 \\ -5 \end{pmatrix}$$

$$= \begin{pmatrix} -5+7\lambda \\ 4\lambda \\ 5-5\lambda \end{pmatrix}$$

Also since $AC \perp BD$:

$$\vec{BN} \cdot \vec{AC} = 0$$

$$\begin{pmatrix} -5+7\lambda \\ 4\lambda \\ 5-5\lambda \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 4 \\ -5 \end{pmatrix} = 0$$

$$-35 + 49\lambda + 16\lambda - 25 + 25\lambda = 0$$

$$90\lambda = 60$$

$$\lambda = \frac{2}{3}$$

$$\text{So, } \vec{BN} = \begin{pmatrix} -5 + \frac{14}{3} \\ \frac{8}{3} \\ 5 - \frac{10}{3} \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} -1 \\ 8 \\ 5 \end{pmatrix}$$

$$\text{And } \vec{OD} = \vec{OB} + 2\vec{BN}$$

$$= \begin{pmatrix} 4 \\ \frac{4}{3} \\ 2 \end{pmatrix} + \frac{2}{3} \begin{pmatrix} -1 \\ 8 \\ 5 \end{pmatrix} = \frac{2}{3} \begin{pmatrix} 5 \\ 10 \\ 8 \end{pmatrix}$$

Question 16

a)

$$\text{For } t = \tan \frac{x}{2} \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1+t^2)$$

$$dx = \frac{2dt}{1+t^2}$$

$$x=0, t=0 \quad \text{and} \quad x=\frac{\pi}{2}, t=1$$

$$\text{LHS} = \int_0^1 \frac{1}{2 - \frac{1-t^2}{1+t^2} + \frac{4t}{1+t^2}} \times \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{2}{2+2t^2-1+t^2+4t} dt$$

$$= \int_0^1 \frac{2}{3t^2+4t+1} dt$$

$$= \int_0^1 \frac{2}{(3t+1)(t+1)} dt$$

$$\text{let } \frac{2}{(3t+1)(t+1)} = \frac{A}{3t+1} + \frac{B}{t+1}$$

$$\text{and } A=3, B=-1$$

$$\text{LHS} = \int_0^1 \left(\frac{3}{3t+1} - \frac{1}{t+1} \right) dt$$

$$= \left[\ln(3t+1) - \ln(t+1) \right]_0^1$$

$$= \ln 4 - \ln 2 - \ln 1 + \ln 1$$

$$= \ln 2$$

$$b) i) \frac{n-1}{2} (1+(n-1))$$

$$= \frac{n(n-1)}{2}$$

$$ii) \text{ Let } f(x) = x - \ln(1+x) \text{ for } x \geq 0$$

$$f(0) = 0 - \ln 1 = 0$$

$$f'(x) = 1 - \frac{1}{1+x}$$

$$= \frac{1+x-1}{1+x}$$

$$= \frac{x}{1+x}$$

$$> 0 \quad \forall x > 0$$

So, $f(x)$ is increasing $\forall x > 0$
and since $f(0) = 0$, $f(x) > 0 \quad \forall x > 0$
 $\therefore x - \ln(1+x) > 0 \quad \forall x > 0$

$$\therefore x > \ln(1+x) \quad \forall x > 0$$

iii) Note:

$$\binom{n}{2} = \frac{n!}{2(n-2)!} = \frac{n(n-1)(n-2)!}{2(n-2)!}$$

$$= \frac{n(n-1)}{2}$$

$$= 1+2+3+\dots+(n-1) \text{ (by (i))}$$

From part (ii):

$$1 > \ln 2$$

$$2 > \ln 3$$

$\vdots$

$$n-1 > \ln n$$

Adding together gives:

$$1+2+3+\dots+(n-1) > \ln 2 + \ln 3 + \dots + \ln n$$

$$\binom{n}{2} > \ln(2 \times 3 \times 4 \times \dots \times n)$$

$$\binom{n}{2} > \ln(n!)$$

$$e^{\binom{n}{2}} > e^{\ln(n!)} \quad \text{since both sides are positive}$$

$$\therefore e^{\binom{n}{2}} > n!$$

c) i) Since $f(x)$ is the sum of squares of reals,
 $f(x) \geq 0 \quad \forall x \in \mathbb{R}$.

$$\left(\sqrt{a_k} x - \frac{1}{\sqrt{a_k}} \right)^2 = (\sqrt{a_k} x)^2 - 2 \frac{\sqrt{a_k} x}{\sqrt{a_k}} + \left(\frac{1}{\sqrt{a_k}} \right)^2$$

$$= a_k x^2 - 2x + \frac{1}{a_k}$$

$$f(x) = \sum_{k=1}^n \left(a_k x^2 - 2x + \frac{1}{a_k} \right)$$

$$= (a_1 + a_2 + \dots + a_n) x^2 - 2n x + \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right)$$

Since $f(x) \geq 0$, $\Delta \leq 0$:

$$(-2n)^2 - 4(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \leq 0$$

$$-4(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \leq -4n^2$$

$$(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \geq n^2$$

ii) Let $a_1=1$, $a_2=2$ etc. (i.e. $a_n=n$)

$$\therefore (1+2+\dots+n)\left(\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\dots+\frac{1}{n}\right) \geq n^2$$

$$\frac{n(1+n)}{2} \left(1+\frac{1}{2}+\frac{1}{3}+\dots+\frac{1}{n}\right) \geq n^2$$

$$1+\frac{1}{2}+\frac{1}{3}+\dots+\frac{1}{n} \geq \frac{2n^2}{n(1+n)}$$

$$\therefore 1+\frac{1}{2}+\frac{1}{3}+\dots+\frac{1}{n} \geq \frac{2n}{n+1}$$